The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science
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MAT 3007 — Optimization

Exercise Sheet 4

Exercise E4.1 ((20pts): Dual Formulation and Weak Duality):
A company produces two products, P_1 and P_2, and aims to maximize profit. The profits per unit are \$50 for P_1 and \$40 for P_2. The production is subject to the following constraints:

$$2x_1 + x_2 \leq 100, \quad x_1 + 3x_2 \leq 90, \quad x_1, x_2 \geq 0,$$

where x_1 and x_2 represent the quantities of P_1 and P_2 produced, respectively.

a) Formulate the dual problem and clearly define its variables and constraints.

b) Solve the dual problem graphically.

c) Verify that the Weak Duality Theorem holds for any feasible solutions of the primal and dual problems.

Exercise E4.2 ((20pts): Strong Duality):
Consider the following primal linear program:

$$\text{minimize } 4x_1 + 5x_2$$

subject to:

$$x_1 + x_2 \geq 4, \quad 2x_1 + 3x_2 \geq 9, \quad x_1, x_2 \geq 0.$$

a) Formulate the dual problem.

b) Solve the dual problem graphically.

c) The optimal solution of the primal is $(x_1^*, x_2^*) = (3, 1)$. Verify that the primal and dual objective values are equal, as required by the Strong Duality Theorem.

Exercise E4.3 ((20pts): Economic Interpretation of Duality):

A farmer needs to purchase fertilizers to maximize the yield of two crops, C_1 and C_2. The farmer can buy two types of fertilizers, F_1 and F_2, with the following nutrient contributions:

Fertilizer	Nutrient A (kg)	Nutrient B (kg)	Cost (\$ per kg)
F_1	3	2	5
F_2	1	4	4

The farmer requires at least 30 kg of Nutrient A and 40 kg of Nutrient B. Let x_1 and x_2 represent the quantities of F_1 and F_2 purchased, respectively.

a) Formulate the primal problem to minimize the total cost of fertilizers while meeting the nutrient requirements.

b) Formulate the dual problem and provide an economic interpretation of the dual variables (e.g., the value of Nutrient A and Nutrient B).

c) Solve both the primal and dual problems using any mathematical software (e.g., Python, MATLAB) and verify that the solutions satisfy the Strong Duality Theorem.

Exercise E4.4 ((20pts): Duality in Real-Life Applications):
A transportation company needs to ship goods from two warehouses (W_1 and W_2) to three retail stores (S_1, S_2, S_3). The supply capacities of W_1 and W_2 are limited to 500 and 400 units, respectively. The demand at S_1, S_2, S_3 is at least 300, 400, and 200 units, respectively. The shipping costs (in $ per unit) are summarized in the table below:

	S_1	S_2	S_3
W_1	6	8	10
W_2	7	5	9

a) Formulate the primal problem to minimize the total transportation cost while meeting supply and demand constraints.

b) Formulate the dual problem and explain the interpretation of the dual variables (e.g., the shadow prices of supply capabilities and demands).

c) Solve the primal and dual problems using any mathematical software (e.g., Python, MATLAB) and verify that the Strong Duality Theorem holds.

Exercise E4.5 (Problem 5 (20pts): Farkas' lemma):
Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Consider the following linear optimization problemÔºöö

$$\text{minimize } \mathbf{b}^\top \mathbf{y}$$

subject to

$$\mathbf{A}^\top \mathbf{y} \geq 0.$$

a) Write the dual problem corresponding to the above minimization problem.

b) Use the Weak and Strong Duality Theorems to show that:

(1). If $\mathbf{b}^\top \mathbf{y} < 0$ for a feasible $\mathbf{y}$, then there cannot exist $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq 0$.

(2). If $\mathbf{b}^\top \mathbf{y} \geq 0$ for all feasible $\mathbf{y}$, then there must exist $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq 0$.

c) Use the results from the above steps to prove Farkas' Lemma.

> Farkas' lemma- Exactly one of the following two assertions is true:
>
> (1). There exists $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq 0$.
>
> (2). There exists $\mathbf{y} \in \mathbb{R}^m$ such that $\mathbf{A}^\top \mathbf{y} \geq 0$ and $\mathbf{b}^\top \mathbf{y} < 0$.

Sheet 4 is due on **Mar, 16th**. Submit your solutions before **Mar, 16th**, **11:00pm**.